

HERMITE FUNCTIONS

Hermite's differential equation

Hermite functions as solutions of the Hermite ODE,

$$H_n''(x) - 2xH_n'(x) + 2nH_n(x) = 0$$

Here n is a parameter. When $n \geq 0$ is integral, this ODE will have a solution $H_n(x)$ which is a polynomial of degree n ; these solutions are known as Hermite polynomials.

In the presence of appropriate boundary conditions, the Hermite ODE is a Sturm Liouville system.

$$H_{2n}(0) = (-1)^n \frac{(2n)!}{n!}, \quad H_{2n+1}(0) = 0, \quad n = 0, 1, \dots$$

Generating function

The Hermite polynomials could be generated from their Rodrigues formula, and that, in turn, a Rodrigues formula can be obtained from the underlying ODE. Also we can go from the Rodrigues formula to a generating function for a given polynomial set:

$$g(x, t) = e^{-t^2 + 2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$$

$$g(-x, -t) = \sum_{n=0}^{\infty} H_n(-x) \frac{(-t)^n}{n!} = g(x, t)$$

The Maclaurin expansion of the generating function

$$e^{-t^2 + 2tx} = \sum_{n=0}^{\infty} \frac{(2tx - t^2)^n}{n!} = 1 + (2tx - t^2) + \dots$$

gives $H_0(x) = 1$ and $H_1(x) = 2x$, and then the recursion formula. Special values of the Hermite polynomials follow from the generating function for $x = 0$:

$$e^{-t^2} = \sum_{n=0}^{\infty} \frac{(-t^2)^n}{n!} = \sum_{n=0}^{\infty} H_n(0) \frac{t^n}{n!},$$

We also obtain from the generating function the important parity relation

$$H_n(x) = (-1)^n H_n(-x)$$

Recurrence Formula

Note the absence of a superscript, which distinguishes Hermite polynomials from the un-related Hankel functions. From the generating function we find that the Hermite polynomials satisfy the recurrence relations

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$$

and

$$H'_n(x) = 2nH_{n-1}(x)$$

$$H'_{n+1}(x) = 2(n+1)H_n(x) = 2H_n(x) + 2xH'_n(x) - H''_n(x)$$

Hermite Polynomials:

$$H_0(x) = 1$$

$$H_1(x) = 2x$$

$$H_2(x) = 4x^2 - 2$$

$$H_3(x) = 8x^3 - 12x$$

$$H_4(x) = 16x^4 - 48x^2 + 12$$

$$H_5(x) = 32x^5 - 160x^3 + 120x$$

$$H_6(x) = 64x^6 - 480x^4 + 720x^2 - 120$$

In general: $H_{(2n+1)}(0) = 0$ And $H_{2n}(0) = (-1)^n \frac{(2n)!}{n!}$

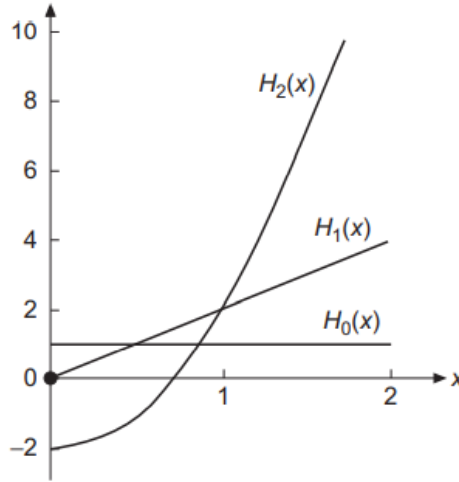
Example:

$$H_5(0) = 0,$$

$$H_7(0) = 0,$$

$$H_4(0) = (-1)^2 \frac{(4)!}{2!} = 12$$

$$H_2(0) = (-1)^1 \frac{(2)!}{1!} = -2$$



Hermite – Polynomial

Rodrigues Formula

A simple way to generate the Rodrigues formula for the Hermite polynomials starts from the observations that

$$g(x, t) = e^{-t^2 + 2tx} = e^{x^2} e^{-(t-x)^2}$$

and

$$\frac{\partial}{\partial t} e^{-(t-x)^2} = -\frac{\partial}{\partial x} e^{-(t-x)^2}$$

We note that n -fold differentiation of the generating function formula, followed by setting $t = 0$, yields

$$\left. \frac{\partial^n}{\partial t^n} g(x, t) \right|_{t=0} = H_n(x),$$

and we can therefore obtain the Rodrigues formula as

$$\begin{aligned} H_n(x) &= \left. \frac{\partial^n}{\partial t^n} g(x, t) \right|_{t=0} = e^{x^2} \left. \frac{\partial^n}{\partial t^n} e^{-(t-x)^2} \right|_{t=0} = \left(-1 \right)^n e^{x^2} \left. \frac{\partial^n}{\partial x^n} e^{-(t-x)^2} \right|_{t=0} \\ H_n(x) &= (-1)^n e^{x^2} \frac{\partial^n}{\partial x^n} e^{-x^2} \end{aligned}$$

Orthogonality

The orthogonality of the Hermite polynomials is demonstrated by identifying them as arising in a Sturm-Liouville system. The Hermite ODE, however, is clearly not self-adjoint, but can be made so by multiplying it by $\exp(-x^2)$. With $\exp(-x^2)$ as a weighting factor,

we obtain the orthogonality integral

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = 0, \quad m \neq n$$

The interval $(-\infty, \infty)$ is chosen to obtain the Hermitian operator boundary conditions.

It is sometimes convenient to absorb the weighting function into the Hermite polynomials. We may define

$$\varphi_n(x) = e^{-x^2/2} H_n(x),$$